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Notes on Classical Electron Dynamics: Lorentz–Dirac Equation

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ABSTRACT

1. THE LORENTZ EQUATION AND THE RUNAWAY PROBLEM

[Conventions, stated once here and used throughout: Gaussian units for electrodynamics, SI for mechanics, $c = 1$. Metric signature and the definition of u^α, a^α to be fixed here. Note that the 2006 text declared Gaussian units but evaluated τ_0 without the factor $1/4\pi\varepsilon_0$; the corrected value is used here from the start.]

1.1. *Electrodynamics does not supply an equation of motion for the charge*

Classical electrodynamics rests on three pillars: the conservation laws, chief among them the conservation of charge; Maxwell's equations; and the Lorentz force law. Maxwell's equations fix the field produced by a given distribution of charge and current. The Lorentz force law fixes the force exerted on a charge by a given field. Between them the two appear to account for everything an electrodynamical problem could ask for, and they do not: neither supplies the equation of motion of the charge itself.

The gap is structural rather than an oversight. To follow the motion of a charge one needs the force acting on it, hence the total field at its location, and the total field includes the field that the charge itself produces. Maxwell's equations say how that self-field is generated but not how it acts back; the Lorentz force law says how a field acts on a charge but is silent on whether the charge's own field belongs among the fields that act. Electrodynamics is complete as an account of how charges make fields and of how fields push charges, and it is incomplete exactly where the two meet on the same particle. Closing that gap is the subject of everything below.

1.2. *Three historical routes*

Any attempt to close the gap must begin by deciding what a charge is, and two pictures divide the subject between them. In the first the charge sits at a point. Its own field then diverges as the point is approached, and with the field the Lorentz force, so the force expression loses its meaning exactly where it is needed. One may object that the Lorentz force was only ever intended for external fields and that the self-field is simply not subject to it; but the objection buys nothing, because it leaves the self-force undetermined and the equation of motion no closer than before. That is the standing difficulty of the point picture. In the second the charge is spread over a finite region and treated as a continuous medium. Every quantity is then finite and the self-force is a well-defined integral over the distribution. The price is that like charges repel, so a body carrying charge of one sign must fly apart; bound charges nevertheless exist, and electrodynamics does not say what holds them together.

Three attempts, spread over thirty years, took these pictures in turn. Poincaré (1906) worked in the extended picture and supplied the missing agent by hand, adding to the electromagnetic stress–energy tensor a binding tensor — the Poincaré stresses — chosen so that the total is in equilibrium and the charged body holds. The construction does what is asked of it, but it postulates a force for the sole purpose of removing a single contradiction, with no other consequence and no independent evidence for its existence; in the century since, few have followed it.

Lorentz (1909) took a position between the two pictures: the electron is a sphere of finite radius carrying a uniform charge. This is concrete enough to yield a definite equation of motion, which is the subject of Sec. 1.3.

Dirac (1938) returned to the point picture, but gave up computing the self-force as the mutual action of one part of the charge on another. He imposed momentum–energy conservation on a thin tube surrounding the world line and read the equation of motion off the balance, so that the divergent self-field enters only through quantities that cancel. This is the route the rest of these notes follow; it is set out in Sec. 2.

These notes adopt the point picture, for the reasons that recommended it to Dirac. It is technically uniform; the model carries no internal structure that has to be specified in advance; the equation of motion comes out in a fixed form; and the construction generalizes, both to curved backgrounds and to the corresponding gravitational problem. The cost is that infinite quantities appear at intermediate stages, so that the force has to be extracted indirectly — either by balancing finite quantities, as in the conservation method, or by separating off a singular part and discarding it. In what sense those operations are legitimate is a question that runs through everything below.

1.3. *The Lorentz model and its three defects*

Lorentz's electron is a sphere of finite radius carrying a uniform charge: far enough into the extended picture that nothing diverges, structureless enough to remain tractable. What holds the charge together is left unanswered, which is precisely the question the Poincaré stresses were invented to settle. In motion the sphere is contracted into an ellipsoid.

Two forces then act on it. One is external. The other is a self-force, produced by the action of the charge elements on one another. At rest this self-force vanishes by symmetry. In motion it does not, and the reason is retardation: the field with which one element acts on a second is the field it carried at an earlier time, and the reaction of the second on the first is delayed in the same way, so the two no longer cancel. The charge elements taken by themselves therefore violate Newton's third law. Nothing is genuinely lost — the missing momentum resides in the field — but the failure is a fair measure of how far the problem has already moved away from mechanics.

1.3.1. *Deriving the self-force directly*

The calculation itself is not deep; the work is all in the expansion. Let the charge be distributed with density ρ over a rigid body of size R, moving slowly enough that magnetic contributions may be dropped to the order kept. The self-force is the double integral of the retarded interaction over all pairs of charge elements,

$$\mathbf{F}_{\text{self}} = \iint d^3r\, d^3r'\, \rho(\mathbf{r})\rho(\mathbf{r}')\, \mathbf{K}(\mathbf{r},\mathbf{r}';\, t-s)\,, \qquad s = \frac{|\mathbf{r}-\mathbf{r}'|}{c}, \tag{1}$$

in which the kernel $\mathbf{K}$ is evaluated not at the present time but one light-crossing time s earlier. Since $s \sim R/c$ is small, expand every retarded quantity about the present instant,

$$f(t-s) = f(t) - s\,\dot{f}(t) + \frac{s^2}{2}\ddot{f}(t) - \frac{s^3}{6}\dddot{f}(t) + \cdots\,, \tag{2}$$

and collect. Each additional power of s costs one factor of R/c and buys one more time derivative of $\mathbf{v}$, so the series organizes itself by derivative order: the term containing $d^n\mathbf{v}/dt^n$ carries R^{n-2}. The zeroth term is the static self-repulsion, which vanishes by symmetry for a rigid symmetric body, and what survives is

$$\mathbf{F}_{\text{self}} = -\frac{4U_0}{3c^2}\,\dot{\mathbf{v}} \;+\; \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}} \;+\; \sum_{n\geq 3} \alpha_n \,\frac{e^2 R^{n-2}}{c^{n+1}}\,\frac{d^n\mathbf{v}}{dt^n}, \tag{3}$$

where U_0 is the electrostatic self-energy and the α_n are pure numbers fixed by the shape of the distribution.

The three groups of terms meet three quite different fates, and the whole subject descends from that split.

The $n = 1$ term carries R^{-1} and diverges as the body shrinks. It is nonetheless harmless in form, being proportional to $\dot{\mathbf{v}}$ and hence indistinguishable from mass times acceleration; it is therefore absorbed,

$$m = m_{\text{bare}} + \frac{4U_0}{3c^2}, \tag{4}$$

which is the first appearance in this subject of what would later be called mass renormalization, and the source of the first defect below.

The $n = 2$ term carries R^0. It is untouched by the point limit, and it is the radiation reaction. Its coefficient is also the one term in the series that cannot depend on how the charge is arranged: with no power of R available, the only quantities left to build a coefficient from are e and c, and the combination is fixed up to a pure number. A sphere, a shell and a dumbbell all give $2e^2/3c^3$, whereas each gives its own coefficient for the divergent term. This is why the radiation-reaction term is believed while the mass term is renormalized away.

The terms with $n \geq 3$ carry positive powers of R and vanish with the body. They are the only place where the structure of the electron would have shown itself, and the point limit destroys exactly that information.

Substituting Eq. (3) into $m_{\text{bare}}\dot{\mathbf{v}} = \mathbf{F} + \mathbf{F}_{\text{self}}$ and using Eq. (4) gives

$$\boxed{m\dot{\mathbf{v}} - \frac{2e^2}{3c^3}\ddot{\mathbf{v}} + \cdots = \mathbf{F}} \tag{AL}$$

now usually called the Abraham–Lorentz equation, in which m is the observed mass and everything from the second term on the left is the self-force.

1.3.2. *The same result from energy balance*

There is a much shorter route, which avoids the charge distribution altogether and therefore never meets the divergent mass. Take the Larmor formula for the radiated power, $P = (2e^2/3c^3)\,\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}$, and demand that the work done by the self-force account for the energy radiated over an interval,

$$\int_{t_1}^{t_2} \mathbf{F}_{\text{self}}\cdot\mathbf{v}\,dt = -\int_{t_1}^{t_2} \frac{2e^2}{3c^3}\,\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}\,dt. \tag{5}$$

Integrating the right-hand side by parts,

$$-\int_{t_1}^{t_2} \frac{2e^2}{3c^3}\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}\,dt = -\frac{2e^2}{3c^3}\Big[\dot{\mathbf{v}}\cdot\mathbf{v}\Big]_{t_1}^{t_2} + \int_{t_1}^{t_2} \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}}\cdot\mathbf{v}\,dt. \tag{6}$$

If the boundary term vanishes — for periodic motion, or whenever $\dot{\mathbf{v}} \cdot \mathbf{v}$ takes the same value at both ends — the two integrands may be compared, and

$$\mathbf{F}_{\mathrm{self}} = \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}}, \tag{7}$$

which is the self-force of Eq. (AL) in three lines. The mass never enters, so it stays where classical mechanics put it.

Two things should be said about what this shortcut costs. First, the conclusion is drawn from equality of integrals over a restricted class of intervals, not from any pointwise identity, so it establishes an average effect and not an instantaneous law; the third and higher derivatives of $\mathbf{v}$, which the direct calculation delivers in Eq. (3), are simply invisible to it. Second, and more interestingly, the term thrown away is not junk. The boundary term $(2e^2/3c^3)\,\dot{\mathbf{v}} \cdot \mathbf{v}$ is the Schott energy, and the piece of the covariant reaction force whose time derivative it is will turn out to be the Schott term. Everything awkward about energy balance in Sec. 3 — where the radiated power and the work done by the reaction force refuse to match instant by instant, and where uniformly accelerated motion radiates while feeling no reaction at all — is already present here, in the term this derivation is allowed to discard because the motion was assumed periodic.

1.3.3. *Three defects*

Three objections can be raised against Eq. (AL).

The mass is entirely electromagnetic, and comes out with the wrong coefficient. The absorbed term gives $m = 4U_0/3c^2$, where U_0 is the electrostatic self-energy, finite here because the radius is. The factor 4/3 is itself a symptom: a covariant accounting of the momentum of a bound charge distribution yields U_0/c^2, and the extra third appears only because the electromagnetic stress has been counted while whatever holds the sphere together has not — the same omission as before, now showing up as a number. Fermi and Wilson later showed that the coefficient becomes unity once the binding stresses are included. The deeper objection survives the repair. If inertia is electromagnetic in origin, a neutral particle should have no mass, and neutral particles have mass. Inertial mass cannot be a manifestation of charge; at most one may say that charge brings mass with it. Nor does Lorentz's derivation claim otherwise on its own terms, since the U_0 appearing in it is electromagnetic energy alone and contains no mechanical rest energy. The energy-balance derivation of Sec. 1.3.2 sidesteps the whole question, which is its chief attraction.

The equation is not covariant. Equation (AL) is written with d/dt in a particular frame. The repair is due to Landau and Lifshitz and is disarmingly short. Replace d/dt by $d/d\tau$ and guess $m\,\dot{u}^\alpha = F^\alpha + k\,\ddot{u}^\alpha$; the guess fails at once, because contracting with u_α and using $u^\alpha u_\alpha = -1$, whose first two derivatives give $u \cdot a = 0$ and $u \cdot \dot{a} = -a \cdot a$, leaves $-k\,(a \cdot a)$ on the right where zero is required. The cure is to add a term along u^α and let the constraint fix its coefficient: demanding $u_\alpha(\dot{a}^\alpha + C\,u^\alpha a \cdot a) = 0$ gives $C = -1$, so that the reaction force must be

$$\frac{2e^2}{3c^3}\left(\dot{a}^\alpha - u^\alpha a^\beta a_\beta\right) = \frac{2e^2}{3c^3}\left(\delta^\alpha{}_\beta + u^\alpha u_\beta\right)\dot{a}^\beta. \tag{8}$$

This is exactly the term that Dirac's far longer construction produces in Sec. 2. Covariance alone, applied to a nonrelativistic result, very nearly hands over the answer — which is worth keeping in mind when judging how much the tube derivation really establishes.

A free electron accelerates itself without bound. This is the serious defect, and it is not repaired anywhere in these notes. It is the subject of Sec. 1.4.

1.4. *The Abraham–Lorentz equation and its runaway solutions*

Set $\mathbf{F} = 0$ in Eq. (AL), keep only the two terms written, and reduce to one dimension. The coefficient of the self-force term is m times a time,

$$\tau_0 = \frac{2e^2}{3mc^3} = 6.27 \times 10^{-24}\ \mathrm{s} \quad \text{(electron)}, \tag{9}$$

which will reappear in every section below, and the equation of motion of a free electron collapses to a first-order equation for the acceleration alone,

$$\dot{a} = \frac{a}{\tau_0}, \qquad \text{whence} \qquad a = C\, e^{t/\tau_0}. \tag{10}$$

Both branches of this solution are unsatisfactory, and for different reasons. If $C \neq 0$, an electron with nothing whatever acting on it accelerates itself, and does so on the timescale τ_0, which is to say it gains a factor of e in acceleration every $\sim 10^{-23}$ seconds. In the nonrelativistic equation the velocity grows without bound; in the covariant version of Sec. 3 it approaches c instead, but the character of the solution is unchanged. No intuition about free particles survives this, and no agent is available to be held responsible for the energy.

If $C = 0$ then $a = 0$, the electron moves uniformly, and Newton's first law is recovered; one may say that the Abraham–Lorentz equation tells us that a free electron moves exactly as an uncharged body does. The physical content is unobjectionable. What is objectionable is the standing of the condition. Equation (10) is first order in a and therefore admits any initial value $a(0)$; the physical solution is picked out by setting $a(0) = 0$, and the only reason offered for doing so is that every other choice misbehaves as $t \to \infty$. A condition imposed at infinity to restrict data at $t = 0$ has no warrant in the initial-value problem it is being applied to, and this remains true no matter how reasonable the surviving solution looks.

The impulse problem sharpens the objection into something concrete. Write the electron's equation with an impulsive force of strength J delivered at $t = 0$,

$$ma - m\tau_0\, \dot{a} = J\, \delta(t), \tag{11}$$

and compare with the corresponding classical problem. A free mass point struck by an impulse jumps discontinuously in velocity and has zero acceleration immediately afterwards: the blow is over, and so is the acceleration. Solving Eq. (11) with the electron moving uniformly beforehand gives instead

$$a = \begin{cases} 0, & t < 0, \\ \kappa\, e^{t/\tau_0}, & t > 0, \end{cases} \qquad \kappa = -\frac{J}{m\tau_0}. \tag{12}$$

The electron does not stop accelerating when the blow ends. It accelerates ever harder, without limit, long after nothing is acting on it, and — since κ carries the opposite sign to J — it runs away in the direction opposite to the blow that started it. This is the runaway, or acceleration collapse, in its starkest form, and no choice of initial data avoids it here: the requirement of uniform motion before the pulse has already been spent.

The root of the behaviour is visible in the order of the equation. Newton's equation of motion contains the acceleration and nothing beyond it, so when the force stops the acceleration stops with it. Equation (AL) contains derivatives of the acceleration, so the state of the electron at a given instant includes how fast its acceleration is changing, and that quantity survives the force that produced it. What sustains it, physically, is the electron's own electromagnetic field. The defect is therefore not an artefact of the sphere model or of the expansion used to reach Eq. (AL); any equation of motion that accounts for the self-field by adding a derivative of the acceleration will inherit it. Dirac's equation, derived on entirely different principles in Sec. 2, inherits it exactly.

1.5. *Criteria for a physically acceptable equation of motion*

Everything from here on consists of proposals to replace Eq. (AL) with something better behaved, and proposals cannot be judged without a standard to judge them by. It is worth fixing that standard now, before any candidate is on the table, so that it cannot be quietly adjusted to fit whichever equation is under discussion. Three requirements are imposed throughout these notes.

1. **The motion must follow from the equation in the ordinary way.** The particle should obey a differential equation that can be solved by the usual procedure — supply initial or boundary data, integrate, obtain the state at all later times — and the data supplied must themselves be data the physical situation could actually provide. This is a weaker demand than it may appear, and Sec. 1.4 has already shown Eq. (AL) failing it: the equation is solvable, but the solution that survives is picked out by a condition imposed at $t \to \infty$, and no experimenter preparing an electron at $t = 0$ has access to that condition.

2. **A stable state must remain stable under a small disturbance.** The concrete test used throughout is the one already applied in Sec. 1.4: take a particle in uniform motion, strike it with a pulse, and ask what it does afterwards. An acceptable equation returns it to uniform motion. It is also acceptable — and this weaker alternative is included deliberately, because Sec. 5 produces solutions of exactly this kind — for the particle to settle into a regular variation confined to a bounded range of speeds. What is not acceptable is for the particle to accelerate without limit toward c, or to swing back and forth between $+c$ and $-c$. Both are read here as signs that the equation has no stable state to return to, rather than as physics.

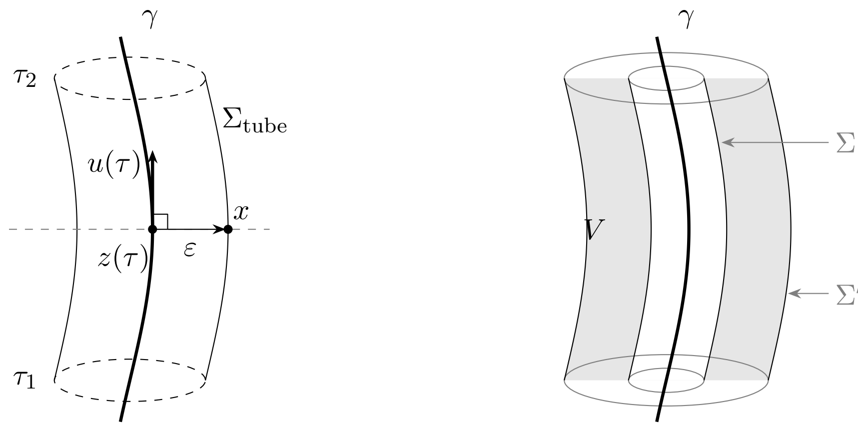
3. **The ordinary physical laws must hold.** Energy conservation and causality are not negotiable, and an equation that purchases good behaviour by breaking either has not solved the problem but relocated it. This criterion does the least work in the pages that follow and is the easiest to forget, which is why it is worth writing down: the sign of a radiation-reaction term, for instance, is exactly the kind of thing one may change to remove a runaway without noticing what it does to the energy budget.

These three are the scorecard. Sections 3, 4 and 5 each close by returning to them. Two features of the list are worth noticing before it is used. It is stated entirely in terms of the behaviour of solutions and says nothing about the form an equation should take, which is deliberate, since the strategy of the sections that follow is precisely to alter the form and inspect what comes out. And it asks that a charged particle behave, in these respects, like an ordinary mechanical one — an expectation that is reasonable, widely shared, and worth flagging as an assumption rather than a deduction.

2. THE LORENTZ–DIRAC EQUATION: DERIVATION

[Terminology: throughout these notes "Lorentz–Dirac" (LD, also called Abraham–Lorentz–Dirac) refers to the classical equation of motion of a point charge, never to Dirac's relativistic wave equation.]

2.1. *The world-tube momentum–energy balance*



(a) (b)

Figure 1. Dirac's construction. *(a)* The tube. At each proper time the point $z(\tau)$ on the world line γ carries a sphere of radius ε, drawn in the hyperplane of simultaneity of $z(\tau)$ — the dashed line — so that the separation $x - z(\tau)$ to a point x of the tube is orthogonal to the four-velocity $u(\tau)$, as in Eq. (15). Dragging that sphere along γ from τ_1 to τ_2 sweeps out Σ_{tube}. The end caps are drawn dashed as a reminder that the integration is carried out over the wall alone; see Sec. 2.5. *(b)* Why the choice of tube does not matter. Two tubes Σ and Σ' enclose the same world line, and the region V between them contains no charge, so the flux of $T^{\mu\nu}$ through the two is the same by Eq. (14). The balance may therefore be read on any surface, including one that keeps well away from the singular world line.

Dirac's advance was a change of question. Lorentz's route, followed in Sec. 1.3, asks what force the charge exerts on itself and answers by summing the retarded action of one charge element on another — a sum that has no meaning for a point charge, since the self-field diverges on the world line. Dirac keeps the point charge and declines to ask about the force at all. He asks instead where the energy and momentum go. Whatever the particle radiates must be paid for out of the particle's own energy and momentum, and that balance can be struck without ever evaluating a field on the world line. The divergent self-field enters the calculation, but only through differences, and the differences are finite.

Surround the world line γ by a three-dimensional tube Σ. The four-momentum crossing it is

$$\Delta P^\mu = \int_\Sigma T^{\mu\nu}\, d\Sigma_\nu. \tag{13}$$

The construction rests on a single observation, and it is worth stating before any tube is chosen, because everything else is bookkeeping. *The answer does not depend on which tube is used.* Let Σ and Σ' both enclose γ and let V be the region between them. That region contains no charge, so the stress–energy tensor is conserved throughout it, and

$$\Delta P'^\mu - \Delta P^\mu = \oint_{\partial V} T^{\mu\nu}\, d\Sigma_\nu = \int_V T^{\mu\nu}{}_{,\nu}\, dV = 0. \tag{14}$$

This is the whole reason the strategy is better than Lorentz's. A force must be evaluated somewhere, and for a point charge every candidate location is a place where the field is infinite. A balance need not be evaluated anywhere in particular: it may be read off on any surface at all, including surfaces that stay a comfortable distance from the singularity, and Eq. (14) guarantees that the reading is the same.

Dirac's tube is the one adapted to the instantaneous rest frame. For a point x on the tube, let $z(\tau)$ be the point of γ satisfying

$$[x^\mu - z^\mu(\tau)]\, u_\mu(\tau) = 0, \qquad [x - z(\tau)] \cdot [x - z(\tau)] = \varepsilon^2, \tag{15}$$

the separation being spacelike. The first condition places x on the hyperplane of simultaneity of $z(\tau)$, so that "at the same moment" means at the same moment for the particle; the second fixes the radius. Together they describe a sphere of radius ε drawn in the instantaneous rest frame and dragged along the world line, sweeping out $\Sigma_{\text{tube}} = \{x(\tau, \varepsilon, \Omega);\ \tau_1 \le \tau \le \tau_2\}$.

Carrying out the integration — the calculation is long, and Poisson's retarded-coordinate tube gives a shorter route to the same place — yields

$$\frac{dP_\mu}{d\tau} = \frac{e^2}{2\varepsilon c^2}\, a_\mu \;-\; e\, f_{\mu\nu} u^\nu, \qquad f^{\mu\nu} = \frac{2e}{3c^3}\left(\dot a^\mu u^\nu - u^\mu \dot a^\nu\right). \tag{16}$$

The second term is finite, and contracting it out with $u \cdot u = -1$ and $u \cdot \dot a = -a \cdot a$ gives $(2e^2/3c^3)(\dot a_\mu - u_\mu\, a \cdot a)$, which is the radiation reaction already met in Eq. (8).

It is worth pausing on the shape of Eq. (16), because it is the same shape that came out of the charged sphere. There, in Eq. (3), the expansion produced a divergent term proportional to the acceleration and a finite term proportional to its derivative. Here the tube integral produces a divergent term proportional to the acceleration and a finite term proportional to its derivative. The two calculations share no assumption worth mentioning — one is an extended body summing its own retarded interactions, the other is a point charge audited for energy and momentum — and they arrive at the same skeleton. That is a good reason to believe the skeleton, and a warning about how much either derivation individually establishes.

One thing Eq. (16) is not, however, is an equation of motion. It records how much four-momentum the *field* carries across the tube. To turn it into a statement about the particle one has to say what the particle's own four-momentum is, and nothing in the calculation supplies that. The freedom left at this point is the subject of Sec. 2.2.

2.2. *The undetermined four-vector B^μ*

[The tube calculation leaves a four-vector B^μ undetermined, restricted only by $v \cdot B = 0$. Dirac's choice is one option among many; this freedom is what Sec. 5 exploits.]

2.3. *The equation*

$$\boxed{ma^\alpha = F^\alpha_{\text{ext}} + \frac{2e^2}{3c^3}\left(\delta^\alpha{}_\beta + u^\alpha u_\beta\right)\dot a^\beta} \tag{LD}$$

[Identify the two pieces of the reaction term: the Schott term and the radiated-momentum term, and note that the projector enforces $u \cdot a = 0$.]

2.4. *The other route: separating the singular part*

[State without proof here that the same equation follows from computing the Lorentz force with $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$ *as the potential. The equivalence of the two routes is proved in Sec. 4.1, and that proof is what opens the door to Sec. 4.]*

2.5. *Two reservations about the derivation*

[As recorded in the 2006 text: (i) the form of B^μ *is arbitrary, fixed only by* $v \cdot B = 0$*; (ii) both derivations evaluate the tube surface integral but not the cap integrals.]*

3. THE LORENTZ–DIRAC EQUATION: DISCUSSION AND SOLUTIONS

3.1. *One-dimensional reduction*

[Parametrize by the rapidity q *via* $\dot{t} = \cosh q$, $\dot{x} = \sinh q$*. Equation (LD) collapses to a single scalar equation:]*

$$\dot{q} - \tfrac{2}{3}\ddot{q} = F^{01}. \tag{17}$$

3.2. *The free solution and the runaway*

$$q = C_1 e^{3\tau/2} + C_2. \tag{18}$$

[$C_1 \neq 0$ *is the runaway;* $C_1 = 0$ *has to be imposed by hand. Note that no initial condition on position and velocity alone selects it — the equation is third order.]*

3.3. *Dirac's asymptotic condition and the price paid*

[Pushing the condition out to $\tau \to \pm\infty$ *eliminates the runaway but introduces pre-acceleration: the particle starts moving before the force arrives, on a timescale* τ_0*. This trades criterion (2) for criterion (3) of Sec. 1.5.]*

3.4. *Energy balance and the Schott term; hyperbolic motion*

[Where the energy sits during acceleration, and the standing puzzle of uniformly accelerated motion, in which the reaction term vanishes although the particle radiates.]

3.5. *Scorecard, and the fork in the road*

[Against the three criteria of Sec. 1.5, LD fails all three in one form or another. Two ways to modify it then present themselves, and they are the subjects of the next two sections:

- sideways — *keep the structure of the reaction term and open up its coefficient (Sec. 4);*
- upward — *keep the coefficient and raise the order of the equation by enlarging* B^μ *(Sec. 5).*

This contrast is the spine of the notes.]

4. SPLITTING OFF THE SINGULAR PART: THE λ–κ FAMILY

4.1. *The equivalence theorem*

[The motion obtained from momentum–energy conservation is the same as the motion obtained by applying the Lorentz force law with $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$*. Give the elementary proof from the 2006 text (Taylor expansion of the retarded potential, term-by-term cancellation), which is more direct than the routes via Zhang Zongsui or Poisson.]*

4.2. *Does the singular part act?*

[Construct the stress–energy tensor of $A_S = \frac{1}{2}(A_{\rm ret} + A_{\rm adv})$ and integrate it. Eq. (1.2.17) of the 2006 text:]

$$\frac{dP^\alpha}{d\tau} = \frac{a^\alpha}{2r} + \frac{2}{3}\dot{a}^\alpha. \tag{19}$$

[The 2006 text read the surviving second term as showing that the singular part is not *inert, and used that as the licence to loosen the split. Flag: this is the load-bearing step of the whole section, and it needs to be checked — see the closing remark of Sec. 4.7.]*

4.3. *The two-parameter family*

$$A^\alpha = \lambda A^\alpha_{\rm ret} + \kappa A^\alpha_{\rm adv}. \tag{20}$$

[Two branches are then singled out by requiring that the retained part have a sensible field-theoretic status.]

4.4. *Branch (a): $\lambda + \kappa = 0$*

[The effective part satisfies the homogeneous Maxwell equations. The free-particle a–v phase portrait has a closed-form solution:]

$$a = \left[-\frac{3}{8\kappa} \ln \frac{1+v}{1-v} + C \right] (1-v^2)^{3/2}. \tag{21}$$

[$\kappa = -1/2$ recovers LD. The resulting equation of motion is Eq. (1.5.1) of the 2006 text:]

$$ma^\alpha = F^\alpha_{\rm ext} - \frac{4\kappa}{3} e^2 \left(\dot{a}^\alpha - a^2 u^\alpha \right). \tag{22}$$

[Behaviour: $\kappa < 0$ gives a convex curve and still runs away; $\kappa > 0$ gives a concave curve that must cross the v axis, so a perturbed particle decelerates *back to uniform motion. The 2006 text called this "Aristotle motion," since force then fixes velocity rather than acceleration.]*

4.5. *Branch (b): $1 - \lambda = -\kappa$*

[Retarded and advanced parts enter the discarded singular term on equal footing. Solve numerically in κ. Result: $\kappa = -1$ is the watershed. For $\kappa > -1$ the runaway survives; for $\kappa < -1$ a critical velocity v_c appears — a particle starting below v_c stays below it, one starting above decelerates stably. Reproduce the table:]

κ	-1.5	-2	-3	-500	-5000
v_c	0.3473	0.2261	0.1341	6.67×10^{-3}	6.67×10^{-4}

4.6. *An unresolved point*

[Extending branch (b) beyond one dimension runs into $\mathbf{n} \cdot \mathbf{v}$ being undefined at the position of the charge itself. The 2006 text suggested the replacement $\mathbf{n} \cdot \mathbf{v} \to \frac{1}{3}v$ but did not carry it out.]

4.7. *What this branch amounts to*

[Eq. (22) is nothing but the LD equation with the radiation-reaction coefficient $2e^2/3$ replaced by a free parameter $-4\kappa e^2/3$. State this plainly — it is the one result in these notes that outlives its original motivation, whatever becomes of Eq. (19).]

5. THE ELIEZER-TYPE SERIES

5.1. *A note on the name*

[Placed first, not in a footnote. The equations of this section follow an ansatz for B^μ taken from Eliezer's 1947 review. They are not *the equation now standardly called the Eliezer, or Eliezer–Ford–O'Connell, equation, which is obtained by order reduction, is free of runaways, and is a live candidate in present-day experiments. Confusing the two inverts the conclusion of this section.]*

5.2. *Expanding B^μ*

$$B_\mu = P_0 v_\mu + P_1 \dot{v}_\mu + P_2 \ddot{v}_\mu + \cdots + P_i v_\mu^{(i)} + \cdots \tag{23}$$

[Differentiating $v^\mu v_\mu = -1$ repeatedly generates a chain of identities whose coefficients form Pascal's triangle; the general term is in Appendix A.4 of the 2006 text. Substituting gives the generalized equation of motion, Eq. (1.3.3) there. Setting all $P_i = 0$ returns LD, Eq. (LD).]

5.3. *Keeping P_2: a third-order equation*

$$\dddot{q} - B\ddot{q} + A\dot{q} - \tfrac{1}{2}\dot{q}^3 = 0, \qquad A = \frac{3m}{2k}, \quad B = \frac{e^2}{k}. \tag{24}$$

[Scan the (A, B) plane numerically. Reported outcome: for A and B of either common sign the particle accelerates to c on its own; for $A, B > 0$ with $A \gg B$ the behaviour is worse still, the velocity swinging at high frequency between $+c$ and $-c$. Figures 1-3-1 through 1-3-4 of the 2006 text.]

5.4. *Keeping P_2 and P_4: a fifth-order equation*

[Eq. (1.3.11) of the 2006 text, scanned over 22 parameter triples (A, B, C). Three classes of behaviour: oscillation between $\pm c$; monotonic approach to c; and a third class, neither reaching c nor returning to uniform motion, in which the speed pulses periodically within a bounded range — "much like an oscillator, except that the oscillator itself keeps moving forward."]

5.5. *Scorecard*

[No parameter choice in either the third- or the fifth-order family satisfies the criteria of Sec. 1.5. Close with the question the 2006 text closed on — whether nature would really select a more complicated B^μ — and leave it open here.]